Chapter 7 · Warehouses, hierarchies and measures

A shop records sales by product, city and date. A manager wants totals by state and month. We are changing the detail of the same measurements. This is the setting for dimensions, hierarchies and aggregation.

Vocabulary. A dimension describes a way to group observations. A measure is the quantity summarized. Grain says what one fact row represents. A hierarchy connects detailed categories to broader categories. A cuboid represents one choice of aggregation level per dimension. The ALL level summarizes a whole dimension.

R33 Schema and grain

Identify the fact table, its grain, dimension references and measures. A star uses comparatively denormalized dimension tables; a snowflake splits dimensions into related tables. Count only what the stated model requires. Joining a fact to a one-to-many dimension relation can duplicate measures, so test multiplicity before summing.

Worked example W33. A fact row Sale(LineID,ProductID,CityID,DayID,Amount) represents one sale line. Product, City and Day are dimensions; Amount is a measure. A star has this fact table plus three flat dimension tables. If Product's category information is moved to a separate Category table referenced by Product, that part becomes snowflaked. Specify grain before aggregating: one sale line is not one daily product total. If product A has a sale of 100 and two tags while B has a sale of 50 and one tag, joining all tags produces 100,100,50. Its sum 250 double-counts A; the true fact sum is 150. A dimension join is safe for summing only when its multiplicity preserves the fact rows.

Check before moving on. Say what one fact row represents. Check join multiplicity before summing a measure.

R34 OLAP operations

Write the grouping before and after the operation. City to State is roll-up; State to City is drill-down. Fixing one dimension value is a slice. Restricting a subcube by several dimension conditions is a dice. Changing the presentation axes is a pivot. Decide from the actual change, not the number of output rows alone.

Worked example W34. A report grouped by State becomes grouped by City: drill-down. City becomes State: roll-up. Keeping only Country='IN' is a slice of the cube. Restricting several product and country values is a dice. Swapping row and column presentation is a pivot; the grouping values and measure can stay unchanged. The operation follows which levels and filters change. More output rows alone do not establish drill-down because a new date range could also add rows.

Check before moving on. Write the before and after grouping levels, filters and presentation. Classify the change from those facts.

R35 CUBE groups and hierarchy counts

Without hierarchies, d dimensions yield 2^d grouping sets. This counts grouping sets, not output rows. If all combinations of n_i values occur, CUBE returns product(n_i+1) groups including totals. For two dimensions with three and two values: 6 detail groups, 3 first-dimension totals, 2 second-dimension totals and 1 grand total, giving 12.

With missing combinations, count distinct groups for each grouping set and add them. For independent linear hierarchies, multiply the available levels in each dimension, including ALL. Branching hierarchies require enumerating the valid level choices under the specified model.

Worked example W35. Facts are (A,IN,10),(B,IN,20),(A,US,30). The Product×Country grouping set has 3 observed groups; Product has 2; Country has 2; the grand total has 1. CUBE returns 8 rows across 4 grouping sets. It does not invent the missing (B,US) detail group. Adding that combination makes the count 4+2+2+1=9. In a separate hierarchy-count question, Geography has City,State,Country,ALL; Product has Item,Category,ALL; Time has Day,Month,Year,ALL. Independent linear level choices give 4×3×4=48 cuboids. Member counts such as 100 cities do not enter this level count. Branching hierarchies need their valid alternatives counted explicitly.

Check before moving on. Are you counting grouping sets, observed rows or hierarchy-level choices? Use the formula for that quantity only.

R36 Measures and correct roll-up

An additive measure can be summed over all relevant dimensions. A semi-additive measure can be summed over some dimensions but not others: an account balance is not generally summed over dates. A non-additive measure needs another aggregation rule.

An average is algebraic: carry sum and count, then divide. An unweighted average of unequal-sized subgroup averages is wrong. Distinct counts and medians generally cannot be recovered from subgroup counts/medians alone. For measure computation questions, identify the quantity and the sufficient summaries before calculating.

Worked example W36. Two sales average 100 and eight sales average 50. Carry sum/count: first group sum 200,count 2; second sum 400,count 8. Combined mean=(200+400)/(2+8)=60. The unweighted average 75 is wrong. SUM,COUNT,MIN,MAX are distributive: summaries can be combined with the same type of aggregation. AVG is algebraic because a fixed-size sum/count summary suffices. Exact MEDIAN is holistic in the general case because a fixed number of subgroup medians cannot determine it. This classification differs from additivity across dimensions: sales amounts are additive; a day's account balances can be summed across accounts but normally not across dates, so balance is semi-additive; a ratio such as margin needs its numerator and denominator rather than summed ratios. Distinct customer counts need overlap information: sets {1,2} and {2,3} have counts 2 and 2 but their union has count 3, not 4.

Check before moving on. Which summaries can actually reconstruct the requested result? Distinguish additive dimensions from distributive/algebraic/holistic computation.

Previous-year questions, worked

Read the original question and options in the linked official paper. These are original explanations, not replacement question statements. PDF page numbers count from the first page of the downloaded file.

DA-2025-Q46 · CUBE output rows

Official question, PDF page 25. Published key: **D**.

The question supplies all combinations of 3 values in one dimension and 2 in the other. Detail grouping contributes 6 rows. The single-dimension groupings contribute 3 and 2; ALL contributes 1. Total 6+3+2+1=12, giving D. There are four grouping sets, not four output rows. The product (3+1)(2+1) works here because all combinations occur.

DA-2026-Q18 · A hierarchy operation

Official question, PDF page 12. Published key: **D**.

The report changes from Country totals to State detail. State is below Country in the hierarchy, so this is drill-down: D. Roll-up would go the other way. Slice/dice restrict values, and pivot changes presentation; neither describes a change to a finer hierarchy level by itself.

DA-2026-Q43 · Count hierarchy cuboids

Official question, PDF page 28. Published key: **A**.

The diagram gives three levels each for Time, Geography and Item. Include ALL as a fourth level in each dimension. Choosing one level independently in each gives 4×4×4=64 cuboids, written 4^3 in option A. Counting only the drawn three levels omits grand aggregation; 2^3 would count binary include/exclude choices rather than the supplied hierarchies.

Chapter practice

Original questions. Write the answer and the intermediate reasoning. Each lettered part belongs to the same problem. The separate solutions guide contains hints, steps and a final check.

P33 · Grain and multiplicity

Sale(LineID,ProductID,CityID,DayID,Amount) has one row per sale line and three flat dimension tables. (a) State the grain and count tables in its star. Move Product's category attributes to a separate Category table: recount. (b) Product A has sale amount 100 and tags red,new; product B has amount 50 and tag red. What is the sale sum before and after joining every tag? Explain the error.

P34 · Classification

Classify: Country→State grouping; City→State grouping; fixing Country='IN'; restricting several Product and Country values; exchanging the report's row and column axes. (b) Does a larger output row count, without knowing the grouping/filter changes, prove drill-down? Give a counterexample.

P35 · CUBE and hierarchies

Facts(Product,Country,Sales)={(A,IN,10),(B,IN,20),(A,US,30)}, with non-null dimensions. (a) Give the number of CUBE grouping sets and total output rows. Add (B,US,40) and recount. (b) Geography has 4 levels including ALL, Product 3, Time 4. How many hierarchy cuboids are possible under independent linear hierarchies?

P36 · Roll-up measures

(a) A group of 2 observations averages 100 and a group of 8 averages 50. Compute the combined mean and name the summaries needed. (b) Customer sets in two groups are {1,2} and {2,3}; find the overall distinct count. Explain why subgroup counts alone are insufficient. Can account balances be summed across accounts and across dates? Classify AVG's computation and balance's additivity.

Before the exam

Start with grain and the requested aggregate. For CUBE, label exactly what is being counted. For roll-up, carry enough information to reconstruct the measure and check for repeated fact rows. Learn both measure classifications; they answer different questions.